

A Novel Sample Clustering Method for Diabetes Data Classification

Quoc Hung Nguyen* [0000-0002-6363-0160], Hoang Huy Nguyen
[0009-0006-9664-264X] and Tu Trinh Nguyen Ngoc
[0009-0000-4591-0583]

*University of Economics Ho Chi Minh City, Vietnam.

*Corresponding author(s). E-mail(s): hungngq@ueh.edu.vn;

Abstract

Recently, patient health records have increased continuously, particularly those related to diabetes. Training models to classify diabetes has created a new opportunity of research for this serious disease; however, there are still limitations related to the accuracy of rare data samples. Furthermore, identifying data samples that support research and education remains a major challenge for AI healthcare researchers. This study proposes a new approach based on intentional data filtering. Specifically, K-Means clustering was used, followed by variable-level noise rules to divide clean and noisy subsets. The clustering method was adjusted using Cohen's D thresholds and classification rules proposed by the team to differentiate between normal and rare valuable samples. Experimental performance has shown the efficiency of the proposed dividing sample strategy. On the normal sample dataset, the Logistic Regression model indicated 91% accuracy, with an F1-score of 0.93 for class 0 and 0.88 for class 1, demonstrating stable and explainable performance. On the rare subset, the Random Forest model got 87% accuracy, with balanced precision and recall averaging 0.87. The "ROC" analysis also supports the stability of the proposed method in handling diverse patient data. This experimental performance confirms that the proposed targeted filtering strategy effectively enhances classification stability in a wide range of medical datasets with numerous valuable samples, demonstrating its practical value for robust medical data analysis and model selection.

Keywords: K-Means, Logistic Regression, Random Forest, Pima Diabetes Dataset, Clustering

1 Introduction

The explosion of technology in the healthcare sector has opened up numerous new approaches for disease diagnosis and treatment [1]. The application of artificial intelligence in medicine has enabled more in-depth, rapid, and efficient analysis and evaluation of patient data [2, 3]. Among these, diabetes is one of the chronic diseases with serious potential complications if not detected and treated early [4, 5]. According to recent statistics, approximately 589 million people aged 20 to 79 worldwide are living with diabetes, accounting for around 11.1% of the adult population [6]. In this context, the trend of using AI tools to support early screening and develop personalized treatment plans is increasingly receiving attention, requiring data analysis and clustering models to achieve high accuracy [1, 2, 7].

Although many studies have achieved significant results in diabetes classification, several limitations remain. Most current models merely classify patients into two groups “diabetic” or “non-diabetic” based on general datasets, without paying attention to differences among patient subgroups according to region, age, gender, or biological characteristics [1, 3, 7, 8]. Model accuracy has improved but remains general and does not reflect the specific characteristics of each sample group. Moreover, deep learning models often require large datasets to perform effectively, whereas the classical Pima Indians Diabetes dataset contains only 768 samples, which is relatively small and prone to bias when encountering outlier samples [1, 7–9]. Previous studies have also tended to remove outliers to increase model stability, inadvertently discarding rare samples that may have high diagnostic value [10, 11].

This study aims to propose a data filtering method to clearly separate common patterns from outliers in a diabetes dataset. Specifically, the main components include:

- Develop a data filtering method to identify and separate frequent patterns from outliers in a diabetes dataset.
- Develop predictive models that are tailored to each sample group, ensuring better learning and generalization in the case of imbalanced data distributions.
- Provides the ability to interpret input features to better understand the role of each feature in predicting diabetes.
- Support analysis of rare specimens, thereby improving the ability to diagnose and personalize treatment.

The primary contributions of this study include:

- Proposed method to cluster data and filter rare and infrequent patterns in diabetes dataset to improve the discrimination ability between different groups of patterns in the same variable.
- Select a more suitable classification model for each post-filtered dataset by partitioning the data into groups of common and outlier samples.
- Providing medical interpretability for outlier samples, supporting clinical diagnosis and in-depth analysis of diabetes.

2 Literature Review

One study delved into comparing traditional machine learning algorithms, in which XGBoost demonstrated superior performance across accuracy, sensitivity, and AUC-ROC metrics. The study analyzed the importance of clinical features such as glucose, BMI, and age, enhancing the interpretability of results [8]. However, the dataset was limited in size and diversity, indicating that the data mainly focused on the majority samples, without specifically analyzing the rare samples.

Recent studies have applied deep learning architectures with attention mechanisms to improve predictive performance on the Pima dataset [9, 12–14]. A notable example is an Attention-Enhanced DNN model, which reported very high results on Pima (98.4% accuracy) and validated on the larger Diabetes-NHANES dataset, demonstrating that “automatic feature weighting” can enhance the distinction of important features in medical data [12]. Nevertheless, such remarkable results on Pima should be interpreted cautiously because Pima is small, and deep models are prone to overfitting without validation on larger datasets.

Another approach combines feature selection with sequential modeling: a study proposed RFE-GRU (Recursive Feature Elimination with GRU) applied on Pima, aiming to remove noisy features before feeding into the GRU classifier; the authors reported significant improvements in accuracy, F1-score, and AUC compared to traditional baselines [15]. Simultaneously, stacking studies (combining multiple models) showed that ensembles combined with RFE/Isolation-Forest can increase stability and reduce the influence of outliers compared to a single model on Pima [16, 17]. These works show that many recent studies apply dimensionality reduction together with multiple models to benefit from the advantages of different approaches.

Several recent studies have explored feature selection methods based on statistical or distributional characteristics. One example is the use of copula-based ranking to identify features that are more closely related to the positive class [18, 19]. Experimental results indicate that this method frequently identifies glucose, BMI, and age as important features. It can also result in modest gains in AUC when used in conjunction with common classifiers. Such approaches prioritize creating models which stay simple and interpretable but still achieve reasonable predictive performance, which is typically favored for clinical use versus a complex black-box model.

In general, accuracy or AUC reports on the Pima dataset suggest that approaches such as attention-based models, RFE-GRU and stacking methods commonly obtain quite higher performance than classical models (LR, KNN and SVM). These methods can remain competitive when evaluated on larger datasets [16]. Recent feature selection methods such as RFE and copula-based filters can reduce the number of features while keeping the models relatively easy to interpret, which is useful when results need to be explained to clinicians. However, some extremely high results reported on the Pima dataset may come from extensive parameter tuning or the use of additional data. Without careful cross-validation, these results may not be fully reliable.

Research Gaps:

- The Pima dataset is relatively small (768 samples) and does not represent diverse populations well. Therefore, validation on multiple populations and the use of long-term follow-up data are necessary.

- Many deep learning and ensemble models are difficult for clinicians to interpret, which limits their practical use in clinical settings.
- Only a few studies clearly specify which model should be used for different patient groups. For instance, one cluster of patients may be better predicted by model X, while another cluster may work better with model Y. Addressing this gap is one of the main goals of the present study.
- Robust clinical trials and external validation studies are still needed before these models can be applied in real clinical practice.

These gaps suggest several practical research directions. One approach is to combine statistically grounded feature selection with simple and interpretable models, and then evaluate the results on large, multi-population datasets with long-term follow-up.

3 The Proposed Model

3.1 Intentional Clustering

The main idea: The Intentional Clustering method was developed to filter noise in medical data based on the discriminative power of variables between two classes. For each variable, outlier values within each class are identified according to their deviation from the general trend; these abnormal values are considered as noise. This process is applied similarly to all variables within the same sample.

We determined the threshold of data cleanliness based on the Cohen’s d index, in which a variable is considered sufficiently clean when its Cohen’s d value is ≥ 0.5 , corresponding to a medium or higher level of discriminative power between classes according to Cohen’s standard [20].

The goal of the Cohen’s D step is to evaluate the ability of a feature to discriminate between classes in a “clean” data set, helping to determine the true difference between groups in the absence of noise. After filtering, the model uses Cohen’s D to test the ability of each original feature to discriminate between classes, thereby evaluating the potential for classification and characterization of the data.

Samples without any confounding variables are conventionally considered common samples, representing typical characteristics of that class (similar to typical characteristics of diseases in reality), while samples with at least one confounding variable are considered rare samples (outliers), serving the ability to explain and reflect the characteristics of abnormal samples in medicine or education. Based on the feature separation of data samples, different machine learning models are applied: common samples are suitable for simple linear models to optimize cost and time, while rare samples can be analyzed with non-linear models such as Random Forest to mine particularly important information. This method simultaneously filters noise and classifies samples according to their level of commonness or rarity, thus optimizing the choice of machine learning models for each subset of data.

Cohen’s D is calculated as follows. General formula:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p} \quad (1)$$

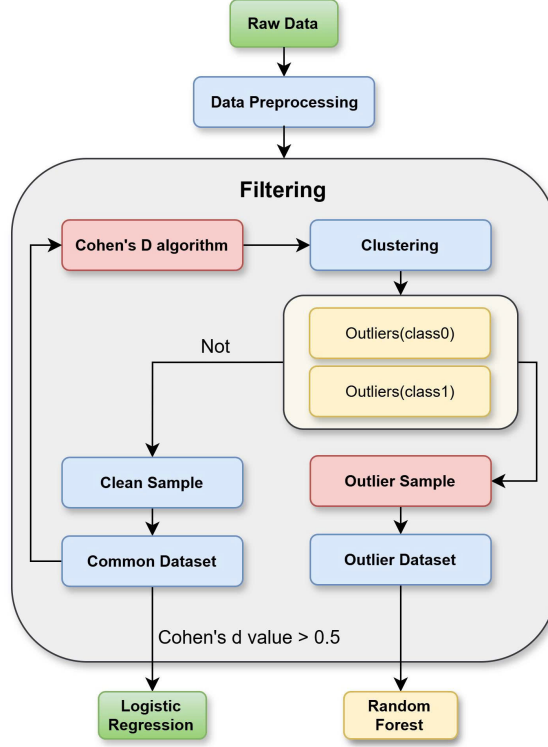


Fig. 1 Chart showing intentional filtering

where $\bar{x}_1$ and $\bar{x}_2$ are the means of the two groups, and s_p is the pooled standard deviation of the two groups, calculated as:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}} \quad (2)$$

where s_1 and s_2 are the standard deviations of each group, and n_1 and n_2 are the sample sizes of the two groups.

3.2 Model with Common Samples

For clean and low-noise data, the research team recommends using simple and lightweight models, such as linear regression, to leverage the clear characteristics of the data without resorting to complex models.

3.3 Model with Outliers

For data with a high level of noise, the research team recommends using models that can effectively adapt to and handle noise, such as random forest. These models

are capable of exploiting information from outlier samples while maintaining stable classification performance, even when the data is heterogeneous or contains abnormal values.

4 Experimental Results

4.1 Dataset

The Pima Indians Diabetes Dataset is often used in studies on diabetes prediction in both medical and machine learning research. It contains medical data collected from female Pima Indian patients aged 21 or older who live in the Phoenix, Arizona area. The dataset includes 768 records, and each record is described by eight numerical features together with a binary label indicating whether the patient has diabetes.

The features included in the dataset are listed below.

- **Pregnancies:** The total number of times the patient has been pregnant.
- **Glucose:** The level of glucose in the blood measured two hours after taking a glucose tolerance test.
- **BloodPressure:** Diastolic blood pressure recorded in millimeters of mercury (mmHg).
- **SkinThickness:** The thickness of the skinfold at the triceps area, measured in millimeters (mm).
- **Insulin:** Serum insulin level measured two hours after the glucose test, expressed in $\mu\text{U/mL}$.
- **BMI:** Body mass index (BMI), defined as body weight (kg) divided by height squared (m^2).
- **DiabetesPedigreeFunction:** A function measuring genetic influence on diabetes risk.
- **Age:** Age of the patient (by years).

The target variable in this dataset is Outcome, which indicates whether a patient has diabetes. A value of 1 denotes the presence of diabetes, whereas 0 indicates that the patient does not have the condition. In the original dataset, some variables contain zero values that are not physiologically plausible. Measurements such as body mass index and blood pressure cannot realistically be zero; therefore, these values are generally treated as missing data. These entries are therefore generally interpreted as missing or unrecorded values rather than valid observations.

During preprocessing, the zero values are replaced with the median of each feature. The Pima dataset is not very large and is commonly used to develop and test classification models.

4.2 Intentional Clustering

Data clustering was conducted using the K-means algorithm, and thresholds were established to identify and filter clean samples. A variable was considered sufficiently clean when its Cohen's d value was ≥ 0.5 , indicating at least a medium level of discriminative power between classes. This process enables validation and assessment

Table 1 Pima Dataset Information Summary

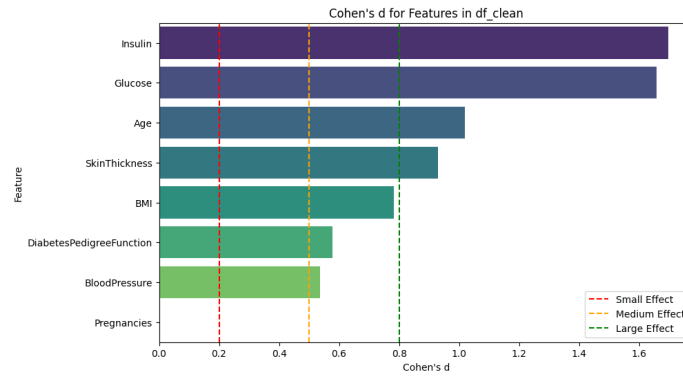
Feature	Description	Data Type
Pregnancies	Number of times pregnant	Integer
Glucose	Plasma glucose concentration	Float
BloodPressure	Diastolic BP (mm Hg)	Float
SkinThickness	Triceps skinfold thickness	Float
Insulin	2-hour serum insulin	Float
BMI	Body mass index	Float
DiabetesPedigreeFunction	Genetic influence measure	Float
Age	Age in years	Integer
Outcome	Diabetes status	Binary

of data cleanliness within each cluster, resulting in final clustering outcomes that clearly reflect differences among sample groups.

Table 2 Number of samples in each dataset after noise filtering through clustering

Data	Quantity
Number of clean samples	446
Number of noisy samples	322

For the dataset containing common samples identified through the Cohen's d algorithm, the results are presented in Fig. 2.

**Fig. 2** Chart illustrating the Cohen's d values of the variables in the common dataset.

4.3 Model Results

4.3.1 Logistic Regression Model

We applied the Grid Search Cross-Validation technique to optimize the model’s hyper-parameters in order to identify the configuration that yields the best predictive performance. The input data were derived from the common dataset, which had been separated in the previous clustering step. The obtained results are presented as follows:

Table 3 Evaluation Results of the Logistic Regression Model

	Precision	Recall	F1-Score	Support
0	0.95	0.91	0.93	58
1	0.85	0.91	0.88	32
Accuracy			0.91	90
Macro Avg	0.90	0.91	0.90	90
Weighted Avg	0.91	0.91	0.91	90

4.3.2 Random Forest

We applied the SMOTE technique to balance the data for the Random Forest model, aiming to address the issue of class imbalance. Subsequently, the model was trained on the outlier dataset, which was extracted from the clustering step used for noise filtering. The obtained results are presented as follows:

Table 4 Evaluation Results of the Random Forest Model

	Precision	Recall	F1-Score	Support
0	0.88	0.86	0.87	86
1	0.87	0.89	0.88	88
Accuracy			0.87	174
Macro Avg	0.87	0.87	0.87	174
Weighted Avg	0.87	0.87	0.87	174

The ROC curve comparing the two models is visualized in Fig. 3.

4.4 Limitations

After denoising and clustering, the dataset size is significantly reduced compared to the original data, which may increase the risk of overfitting in some classification models.

In addition, because the models are trained on filtered subsets rather than the complete dataset, their generalizability to a larger population may be limited. Further validation on larger and more diverse datasets is necessary to address this issue.

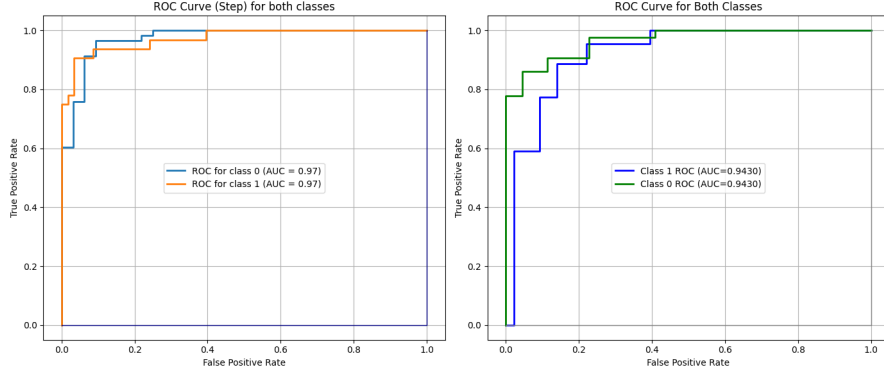


Fig. 3 ROC curve comparison between Logistic Regression and Random Forest models.

5 Conclusion

This study presents a medical data filtering method that can be applied in both medical education and predictive modeling research. The goal is to improve model evaluation results without using common approaches such as dimensionality reduction or removing many samples from the dataset.

The method uses the Cohen's d index together with a clustering process to identify and filter noisy samples. By doing this, typical patient patterns can be separated from rare outliers, making the dataset more suitable for both simple linear models and more complex models that are able to handle noisy data.

In addition, the research team also proposed a proposal to help minimize the impact of noise and reduce misclassification of disease samples (recall) while still retaining rare and infrequent samples, containing valuable information in the medical field, especially for clinical studies with limited patient data.

For future work, the research plans to extend the experiments to more noisy medical datasets; evaluate the performance on various models to verify the generalizability and stability of the proposed method.

However, the proposal still has some limitations. After denoising and clustering, the size of the subset dataset is significantly reduced compared to the original dataset, increasing the risk of overfitting for some models. In addition, because the models are trained on data that is not representative of the entire sample, they may not be fully representative of a larger population, suggesting the need for future research to address this issue.

Acknowledgements. This research is funded by the University of Economics Ho Chi Minh City, Vietnam (UEH).

References

- [1] Okwor, I.A., Hitch, G., Hakkim, S., Akbar, S., Sookhoo, D., Kainesie, J.: Digital technologies impact on healthcare delivery: a systematic review of artificial intelligence (ai) and machine-learning (ml) adoption, challenges, and opportunities. *AI* **5**(4), 1918–1941 (2024)

- [2] Bhuyan, S.S., *et al.*: Generative artificial intelligence use in healthcare: opportunities for clinical excellence and administrative efficiency. *Journal of medical systems* **49**(1), 10 (2025)
- [3] Mizna, S., Arora, S., Saluja, P., Das, G., Alanesi, W.A.: An analytic research and review of the literature on practice of artificial intelligence in healthcare. *European Journal of Medical Research* **30**(1), 382 (2025)
- [4] Migdalis, I.N.: Chronic Complications of Diabetes: Prevalence, Prevention, and Management vol. 13, p. 7001. MDPI, ??? (2024)
- [5] Guan, H., *et al.*: Advances in secondary prevention mechanisms of macrovascular complications in type 2 diabetes mellitus patients: a comprehensive review. *European Journal of Medical Research* **29**(1), 152 (2024)
- [6] International Diabetes Federation: IDF Diabetes Atlas, 11th Edition. <https://idf.org/about-diabetes/diabetes-facts-figures>. Accessed: Oct. 20, 2025 (2025)
- [7] Wang, S.C., Nickel, G., Venkatesh, K.P., Raza, M.M., Kvedar, J.C.: Ai-based diabetes care: risk prediction models and implementation concerns. *NPJ digital medicine* **7**(1), 36 (2024)
- [8] Zhang, Z.: Comparison of machine learning models for predicting type 2 diabetes risk using the pima indians diabetes dataset. *Journal of Innovations in Medical Research* **4**(1), 65–71 (2025)
- [9] Qi, X., Lu, Y., Shi, Y., Qi, H., Ren, L.: A deep neural network prediction method for diabetes based on kendall’s correlation coefficient and attention mechanism. *Plos one* **19**(7), 0306090 (2024)
- [10] Rolando, M., Raggio, V., Naya, H., Spangenberg, L., Cagnina, L.: A labeled medical records corpus for the timely detection of rare diseases using machine learning approaches. *Scientific Reports* **15**(1), 6932 (2025)
- [11] Feng, X., Cai, Y., Xin, R.: Optimizing diabetes classification with a machine learning-based framework. *BMC bioinformatics* **24**(1), 428 (2023)
- [12] Hu, H., Dong, W., Yu, J., Guan, S., Zhu, X.: Utilizing attention-enhanced deep neural networks for large-scale preliminary diabetes screening in population health data. *Electronics* **13**(21), 4177 (2024)
- [13] Zhao, J., Gao, H., Yang, C., An, T., Kuang, Z., Shi, L.: Attention-oriented cnn method for type 2 diabetes prediction. *Applied Sciences* **14**(10), 3989 (2024)
- [14] García-Ordás, M.T., Benavides, C., Benítez-Andrades, J.A., Alaiz-Moretón, H., García-Rodríguez, I.: Diabetes detection using deep learning techniques with oversampling and feature augmentation. *Computer Methods and Programs in*

Biomedicine **202**, 105968 (2021)

- [15] Shams, M.Y., Tarek, Z., Elshewey, A.M.: A novel rfe-gru model for diabetes classification using pima indian dataset. Scientific reports **15**(1), 982 (2025)
- [16] Reza, M.S., Amin, R., Yasmin, R., Kulsum, W., Ruhi, S.: Improving diabetes disease patients classification using stacking ensemble method with pima and local healthcare data. Heliyon **10**(2) (2024)
- [17] Idris, N.F., Ismail, M.A., Jaya, M.I.M., Ibrahim, A.O., Abulfaraj, A.W., Binzagr, F.: Stacking with recursive feature elimination-isolation forest for classification of diabetes mellitus. Plos one **19**(5), 0302595 (2024)
- [18] Aich, A., Murshed, M.M., Hewage, S., Mayeaux, A.: Copulasmote: A copula-based oversampling approach for imbalanced classification in diabetes prediction. arXiv preprint arXiv:2506.17326 (2025)
- [19] Liu, C., Yang, H., Yang, J., Wang, H.: Correlation analysis of diabetes based on copula. Frontiers in Endocrinology **15**, 1291895 (2024)
- [20] Cohen, J.: Statistical Power Analysis for the Behavioral Sciences. Routledge, ??? (2013)